
Copiloting Creative 3D Scene Modeling and Visualization with Generative Agents

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1 Introduction

Creating 3D scenes has been an important task in digital content creation with applications including interior designs, game design, visual effects, product marketing, etc. Recent advancements in generative AI have brought potential improvements to 3D scene creators, as significant progress has been made in AI-assisted automated design and 3D content generation [1, 2, 3]. New developments in AI-assisted 3D content generation include text-to-3D mesh generation [4], procedural model generation [5], and room layout generation [6]. However, although current systems can assist 3D content creators in labor-intensive tasks, few research studies have been conducted to improve 3D creation workflow and inspire idea explorations [7]. In this work, we present a novel copilot system that leverages language and image-generative agents to allow high-quality and iterative 3D scene modeling and visualization and accelerate the creation of creative 3D content. Compared to traditional 3D modeling and visualization workflows, our system provides more natural and intuitive control and can significantly reduce the time required to create 3D content.

Our copilot system is designed to tackle the following key challenges in 3D modeling and visualization: (i) Traditional 3D editing software typically has *complex user interfaces* that introduces a steep learning curve for content creators. (ii) 3D scene modeling and visualization are time-consuming processes that typically require labor-intensive *manual adjustments and fine-tuning* of 3D objects. (iii) Generating high-quality, especially large-scale, 3D scenes can be computationally intensive and time-consuming, making iterative evaluating and refining of 3D scene design difficult.

We summarize our key technical contributions as follows: (i) we design and implement an easy-to-use copilot system for 3D scene modeling and visualization by leveraging text and image generative agents. Our copilot system allows content creators to control the generation from simple and intuitive inputs, such as natural language text and graphical UI. (ii) we develop an LLM-based scene generation agent that leverages a multi-step reasoning workflow to allow high-quality 3D scene creation without labor intensive manual adjustments or fine-tuning. And (iii) we develop a scene subdivision algorithm to allow our scene generation agent to create open-set and large-scale 3D scenes.

2 System Design

System overview. Figure 1 shows an overview of the creation workflow of our system. Our system consists of three components: (i) a web-based user interface, (ii) an LLM-based scene generation agent, and (iii) a diffusion model-based rendering agent. Our user interface offers user-friendly 3D scene editing operations, such as asset importing, object manipulation, and layout adjustment. Our LLM-based scene generation agent, the main component in our system, can take natural language instructions from users and generate high-quality 3D scenes. Notably, our scene generation agent can create open-set and large-scale 3D scenes that do not require domain-specific data at any stage. To render a 3D scene, users can trigger the diffusion model-based rendering agent from the current camera view and add any text descriptions as additional guidance to the diffusion model. Unlike traditional rendering workflows, which require highly detailed 3D assets to produce high-quality results, our system leverages a diffusion model-based rendering agent from the Adobe Firefly API [8] to produce high-quality rendering from simple scene setups with basic object geometries.

*Work partially done while the author interned at Adobe Research.

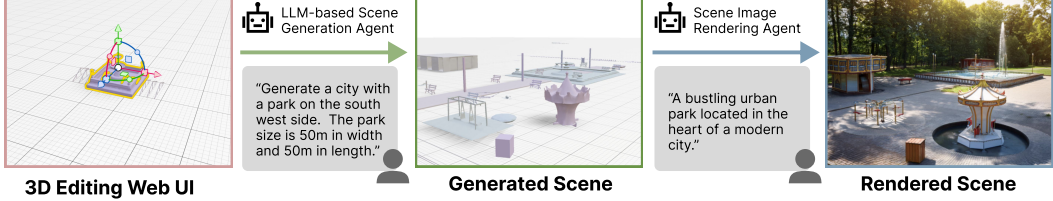


Figure 1: Creation workflow overview. Our system utilizes a web-based interface for 3D scene and object editing. We design an LLM-based scene generation agent to allow easy creation of complicated scenes. The generated 3D scene can then be rendered with a diffusion model-based rendering agent.

Scene generation LLM agent. Recent advancements have allowed LLMs to understand complex natural language inputs and generate structured outputs that follow rigorous rules [9]. Leveraging this key capability of LLM, recent works introduce several systems to allow 3D scenes, represented as a scene graph text format, generated from the user’s natural language descriptions [10, 5, 11]. However, existing system designs often face quality and generalization issues in real-world applications as they rely heavily on domain-specific data for fine-tuning or in-context examples to ensure high-quality outputs. To address this issue, we design a *multi-step reasoning workflow* that utilizes LLM’s built-in knowledge to create high-quality open-set 3D scenes. Specifically, our multi-step reasoning workflow explicitly instructs LLM to plan the scene object positions, orientations, and relative spatial relationships. The generation process can also be configured with multiple conditions, such as object types, number of objects, scene sizes, etc. To scale up the generation to large-sized scenes with many objects, we introduce a scene subdivision algorithm that helps the generation agent break down large generation goals into smaller ones. This design helps the agent to maintain the reasoning complexity and achieve overall high-generation quality. Specifically, the scene subdivision algorithm first generates high-level planning for a scene, such as a city, and then divides the scene into sub-regions, e.g., a park, based on the region sizes. During each subdivision, the algorithm will create a context-inheriting prompt to combine the previous generation’s goal with the current one.

Using the multi-step reasoning workflow and the scene subdivision algorithm, our scene generation agent will output a scene graph. This scene graph includes object category information and detailed spatial information such as sizing and positions. Our generation agent can also create object orientation angles, which current scene generation agents often struggle to do. In the subsequent step, 3D model assets will be retrieved and composited into the scene in the web interface. We implement the retrieval in two steps: (i) CLIP score [12]-based asset matching between asset name and its rendered image and (ii) Object morphology front identification via visual prompting. In the second step, our system will render a set of four object renderings and prompt multi-modal LLM to identify its morphology front so that it correctly aligns with the planned object orientation in the original scene graph. In summary, our scene-generation agent can generate high-quality and open-set scenes ranging from smaller ones, such as living rooms, to large ones, such as a city.

Evaluation. We first show example renderings of a living room and a bedroom in Figure 2. Our system can generate indoor rooms with visually coherent layouts, including detailed object position and orientation. Note that our scene-generation agent generates scene objects with respect to their real-world physical sizes. The agent also ensures that objects are placed in contextually appropriate locations, such as arranging furniture according to common room design principles and maintaining functional flow.

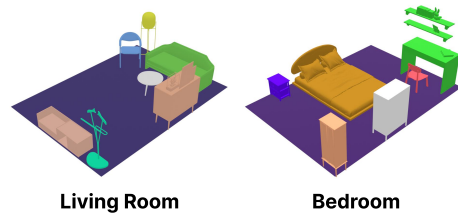


Figure 2: Visualization of indoor rooms generated by our system.

For quantitative evaluation, we evaluate our LLM scene-generation agent on the indoor room generation task against LayoutGPT [10], a recent LLM-based scene-generation system, and ATISS [6], an auto-regressive scene-generation model. We evaluate our system on the 3D-FRONT dataset [13] following LayoutGPT and ATISS. We measure the generation quality regarding the FID scores [14] on the rendered images of the generated scenes. Our system achieves FID scores of 27.51 and 76.21 on generated bedrooms and living rooms, representing a 5.8% and 9.4% improvement over LayoutGPT and ATISS, respectively. Additional results are included in the appendix at Table 1.

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A Appendix

A.1 Subdivision Process Visualization

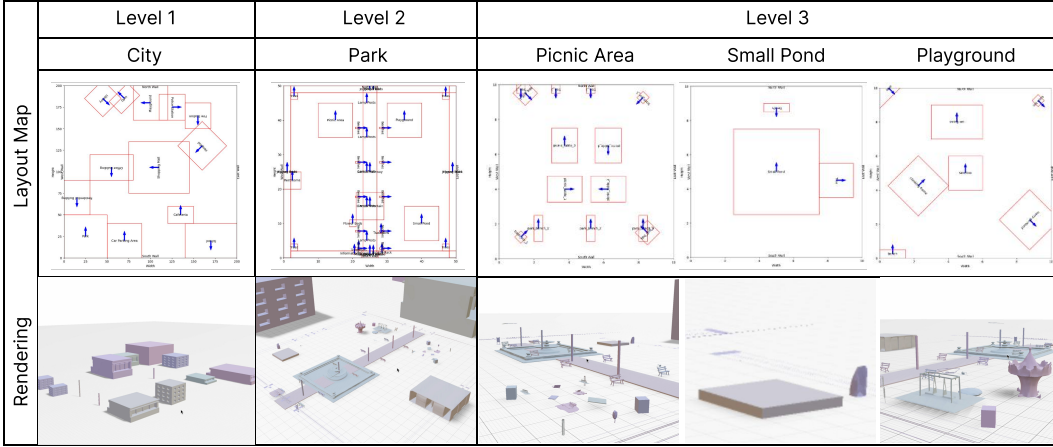


Figure 3: Visualization of the subdivision process in scene generation. Our scene generation agent utilizes a subdivision process to generate high-quality and large-scale scenes. In this example, the entire scene is generated from a single user prompt “Generate a city with a park on the south west side. The park size is 50m in width and 50m in length.” Our scene generation agent first generates a layout for the city and then subdivides the park into a sub-area to generate. Finally, individual function areas in the park are generated by continuing the subdivision. The subdivision process is determined by each area’s size.

A.2 Scene Creation and Rendering Results

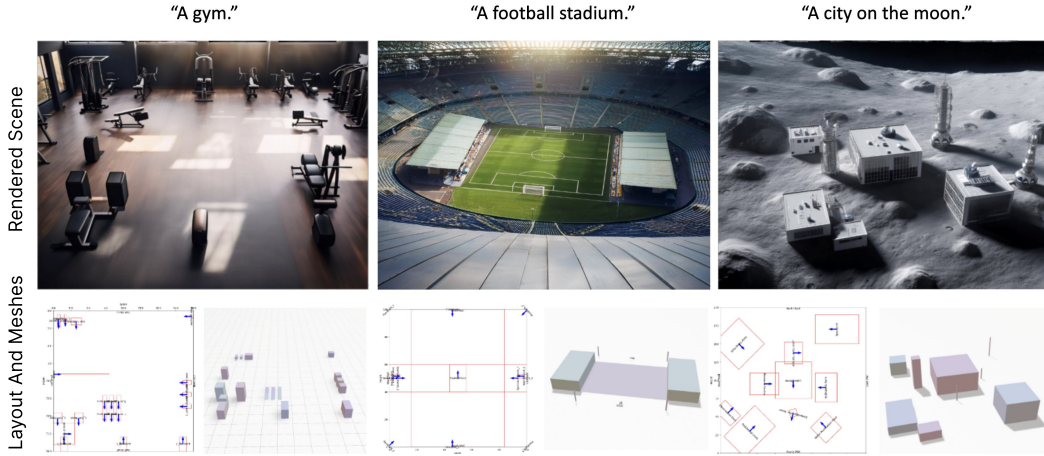


Figure 4: Examples of generated 3D scenes and their renderings. We show three additional 3D scene generation results (row two) with their layout maps (left) and scene object geometries (right). In row 1, we show the scene rendering results created by our diffusion-based rendering models. These results highlight that our system can create open-set 3D scenes of high quality and inspire creative 3D editing and idea exploration.

A.3 Additional Evaluation Results

Table 1: Layout generation quality evaluation.

Models	Bedroom (FID ↓)	Living Room ↓
ATISS	30.02	85.40
LayoutGPT (GPT-3.5)	28.37	76.34
LayoutGPT (GPT-3.5, chat)	29.66	89.40
LayoutGPT (GPT-4)	29.88	78.60
Ours (GPT-4)	27.51	76.21
Ours (GPT-4o)	27.92	75.11